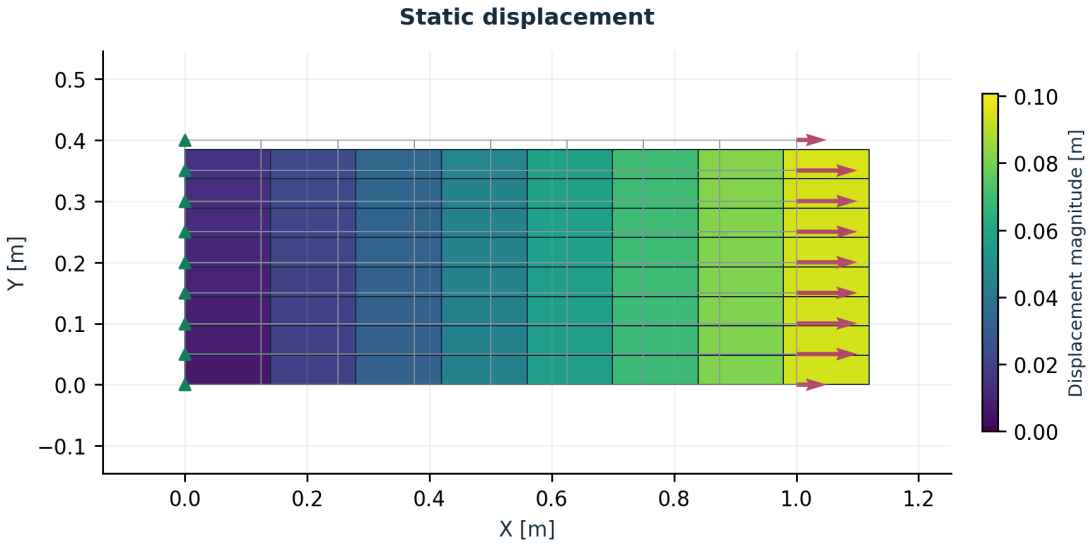
Finite-strain membrane: homogeneous uniaxial tension

ACA05 | SOL 106 | Independent analytical equation verification

A flat membrane in homogeneous tension tests finite-strain equilibrium and Poisson contraction. The reduced plane-stress St Venant-Kirchhoff equations give an exact nonlinear force-stretch relation.

Geometry / mesh	Membrane 1 m by 0.4 m, reference thickness 0.01 m; meshes 1x1, 2x2, 4x4 and 8x8.
Material / section	Synthetic isotropic elastic material: E = 70 MPa, nu = 0.3. Green-Lagrange membrane energy; thickness strain condensed by plane stress.
Boundary conditions	Every node uz=rx=ry=rz=0; ux=0 at x=0; origin uy=0. Other in-plane displacements free.
Applied load	Fixed-direction edge traction for stretches 1.005, 1.02, 1.05 and 1.10; ten continuation steps per independent analysis.



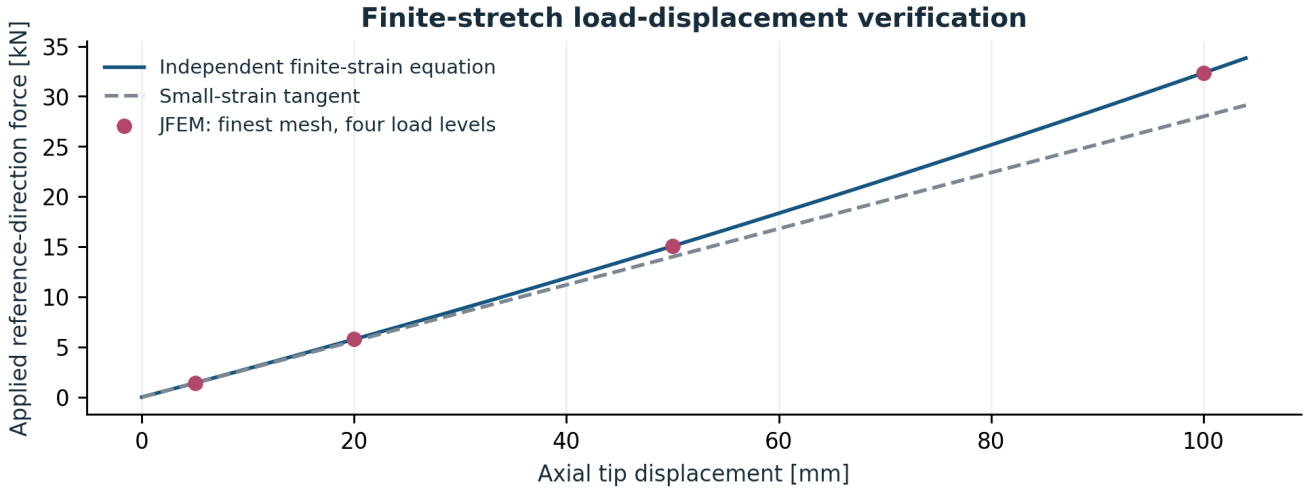
Shape amplified x1.19; grey = original.

Actual solver output. Deformation is amplified for visibility; mode vectors have arbitrary sign and amplitude.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
tip axial displacement [m]	0.1	0.1	9.992e-13%	PASS
top lateral contraction [m]	-0.012805	-0.012805	2.3301e-12%	PASS

Evidence, convergence and interpretation



Independent reference equations

F_2D=diag(lambda,mu); E_GL,2D=(F_2D^T*F_2D-I_2D)/2; thickness strain condensed by plane stress

S_yy=0 implies mu=sqrt(1-nu*(lambda^2-1))

P_xx=lambda*S_xx=E*lambda*(lambda^2-1)/2

F_total=P_xx*b*t; u_tip=(lambda-1)*L

All 16 solves reproduce exact axial displacement and lateral contraction within 5.7e-12%.

At 10% elongation the force is 32340 N, 15.5% above the linearized prediction.

Scope: Homogeneous reduced plane-stress membrane verification of geometric nonlinearity; no finite-rotation bending, plasticity or experimental constitutive validation.

Reference and reproduction

Idaho National Laboratory MOOSE: ComputeStVenantKirchhoffStress
https://mooseframework.inl.gov/source/materials/lagrangian/ComputeStVenantKirchhoffStress.html

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997cce40 | Full precision values: results.json / metrics.csv